Data Science
Data Visualization

Presented By : Harsh Dalal

We Make You Trust Data

Agenda

- Explain what data visualization is and its importance in our world today
- Understand why Python is considered one of the best data visualization tools Describe matplotlib and its data visualization features in Python
- List the types of plots and the steps involved in creating these plots

Data Visualization

Data visualization is a technique to present the data in a pictorial or graphical format.

Well, you might wonder why data visualization is important?

Data Visualization

You are a Sales Manager in a leading global organization. The organization plans to study the sales details of each product across all regions and countries. This is to identify the product which has the highest sales in a particular region and up the production. This research will enable the organization to increase the manufacturing of that product in that particular region.

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Data Visualization

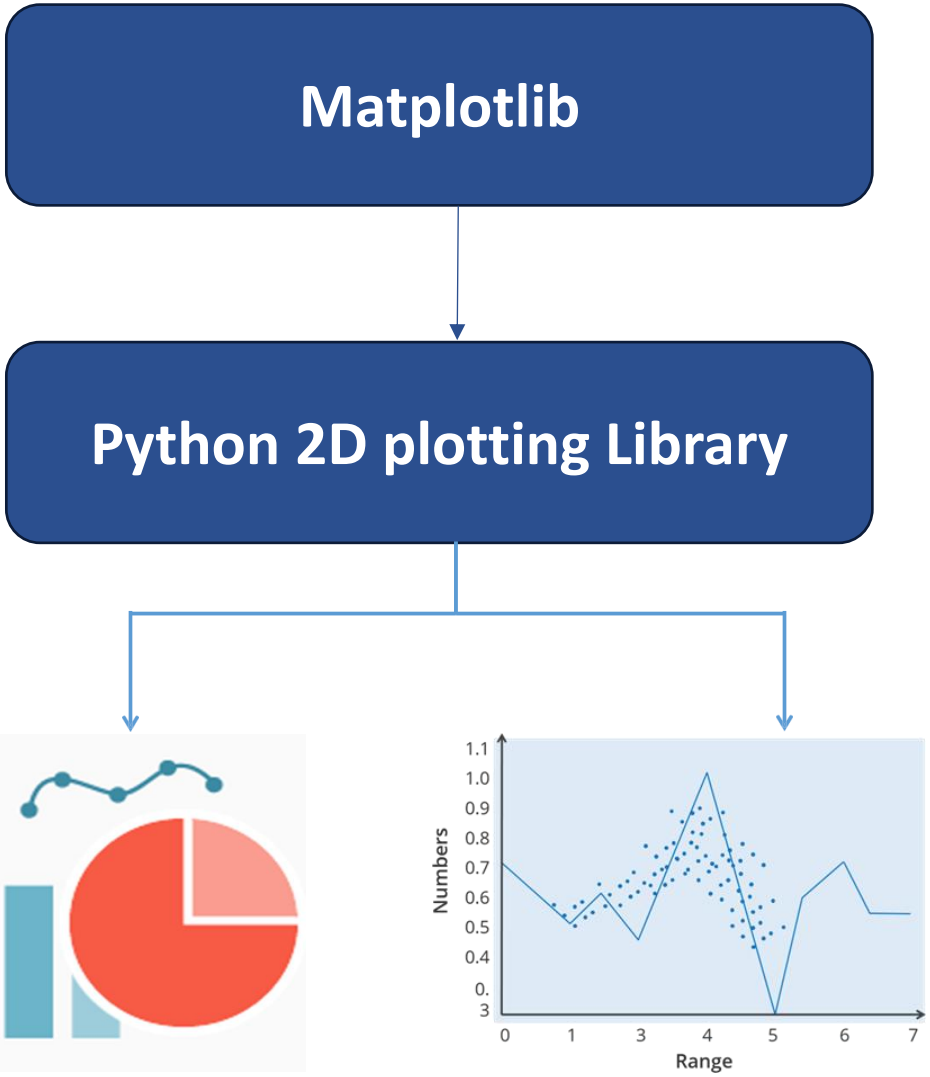
The main benefits of data visualization are as follows:

Data Visualization Considerations

- Three major considerations for data visualization:

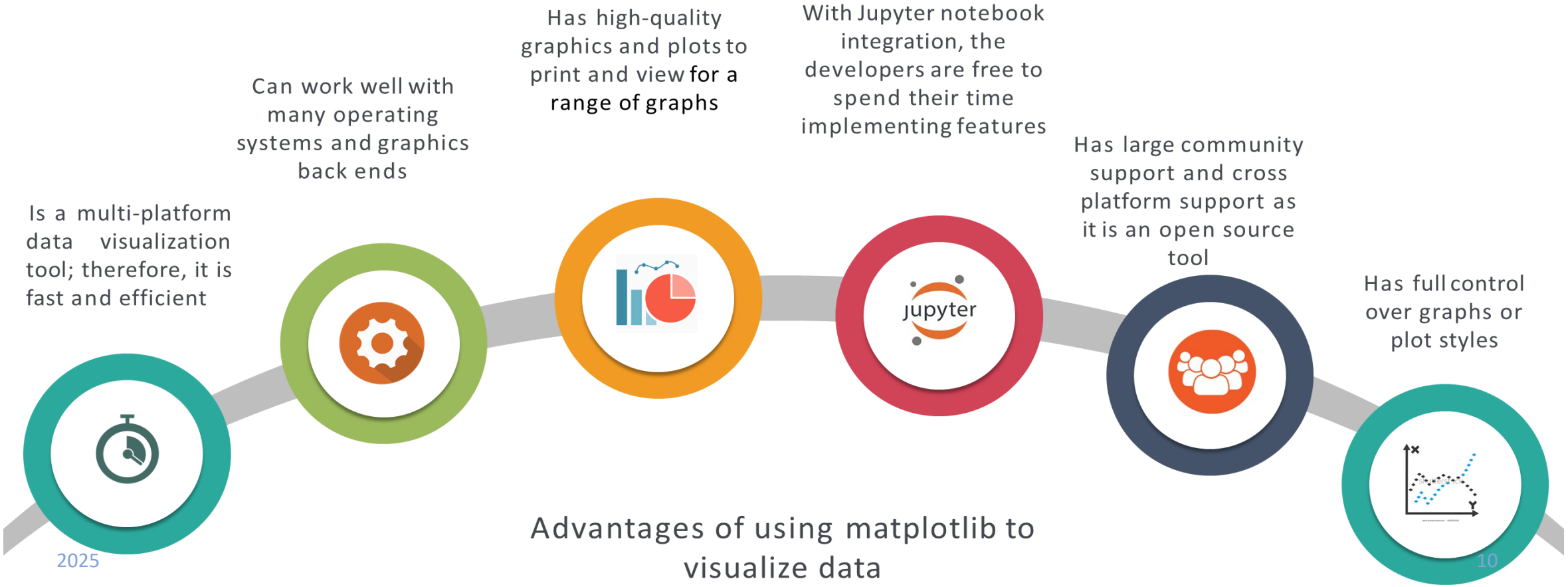
Python Libraries—matplotlib

- Using Python's matplotlib, the data visualization of large and complex data becomes easy.



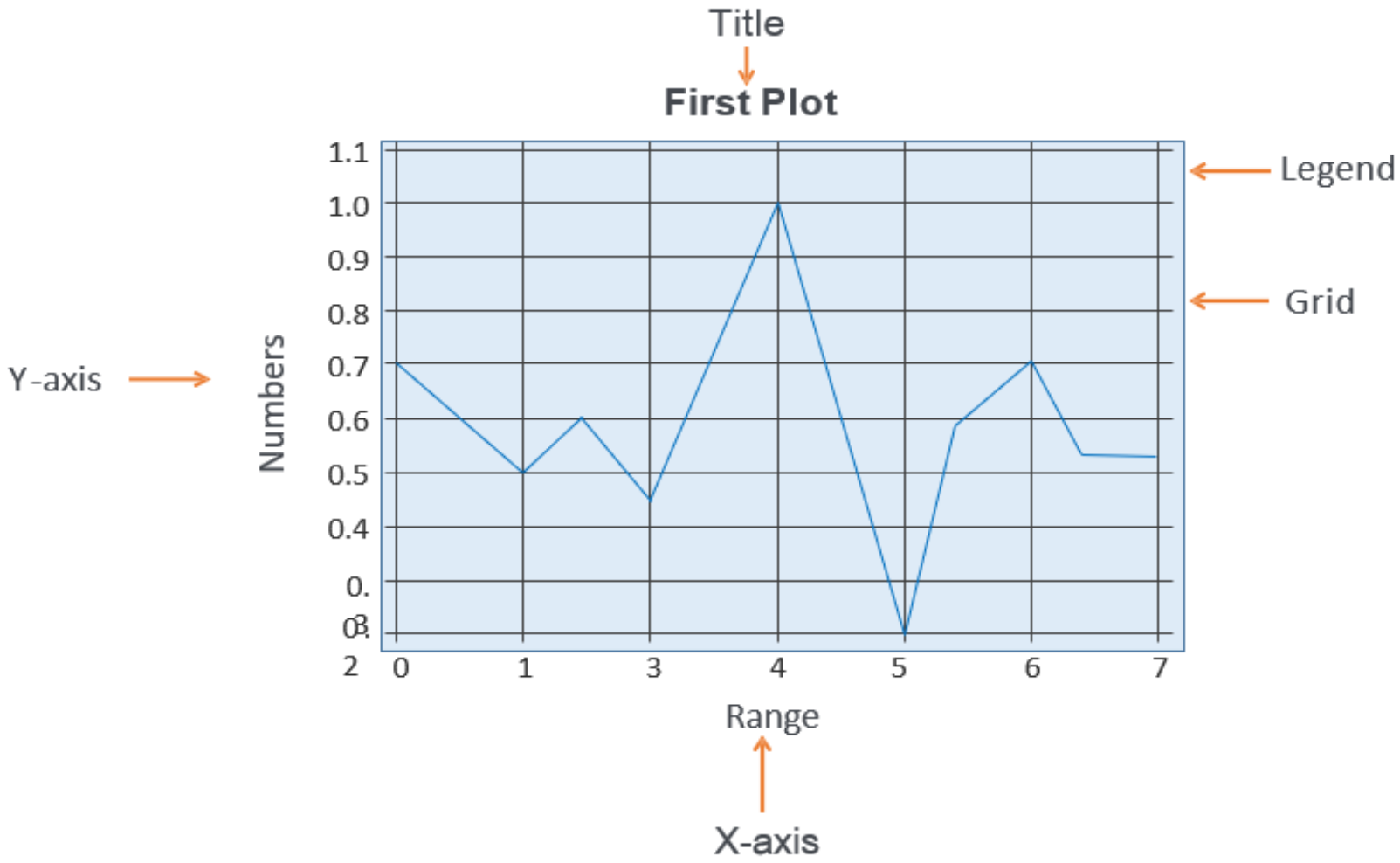
Python Libraries—matplotlib

- There are several advantages of using matplotlib to visualize data. They are as follows:



The Plot

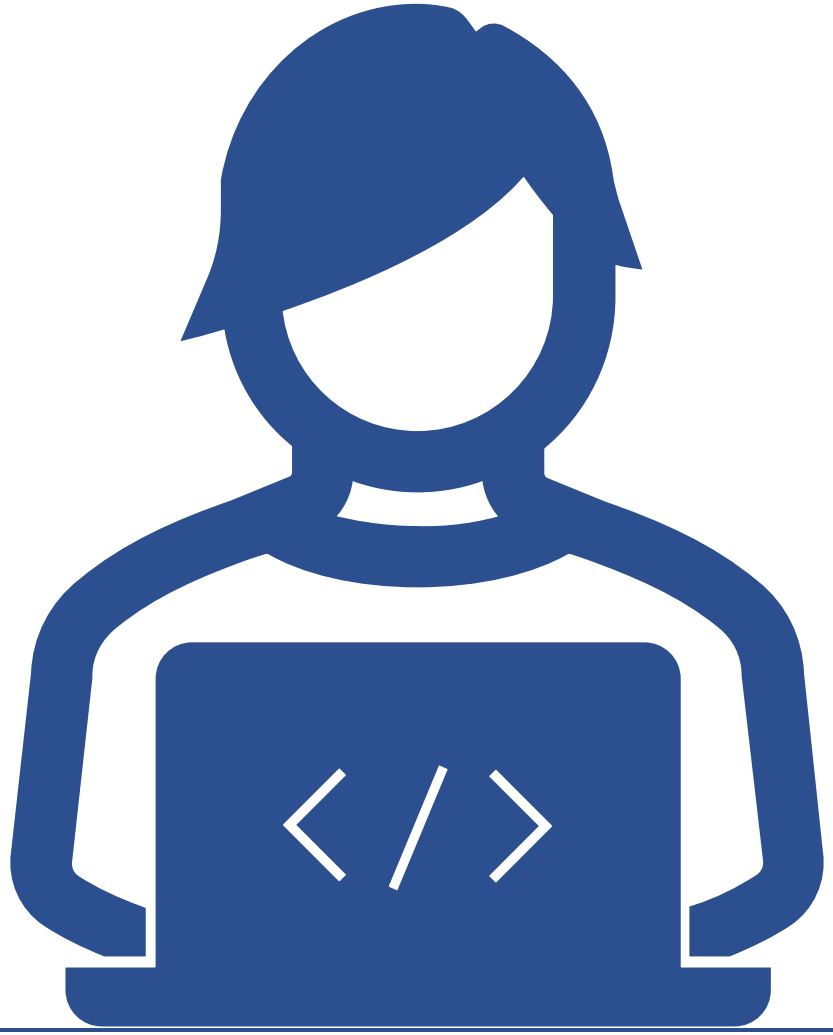
- A plot is a graphical representation of data, which shows the relationship between two variables or the distribution of data.



Steps to Create a Plot

- You can create a plot using four simple steps.

Hands - on



Knowledge Check

- Matplotlib Quiz

Knowledge Check

Which of the following methods is used to set the title?

a. Plot()

b. Plt.title()

c. Plot.title()

d. Title()

Knowledge Check

Which of the following methods is used to set the title?

a. Plot()

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c. Plot.title()

d. Title()

The correct answer is : **b**

Explanation: Plt.title() is used to set the title.

Subplots

- Subplots are used to display multiple plots in the same window.
- With subplot, you can arrange plots in a regular grid.

The syntax for subplot is

subplot(m,n,p) → It divides the current window into an m-by-n grid and creates an axis for a subplot in the position specified by p.

For example,
subplot(2,1,2) creates two subplots which are stacked vertically on a grid. subplot(2,1,4) creates four subplots in one window.

Subplots

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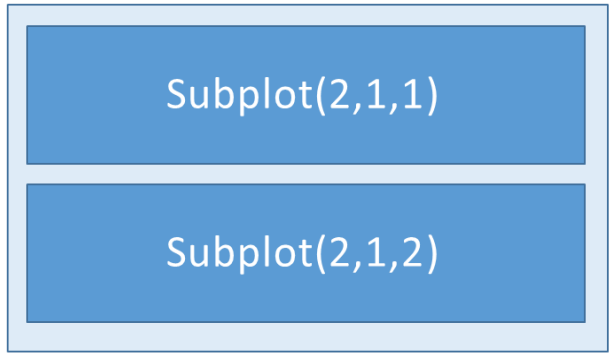
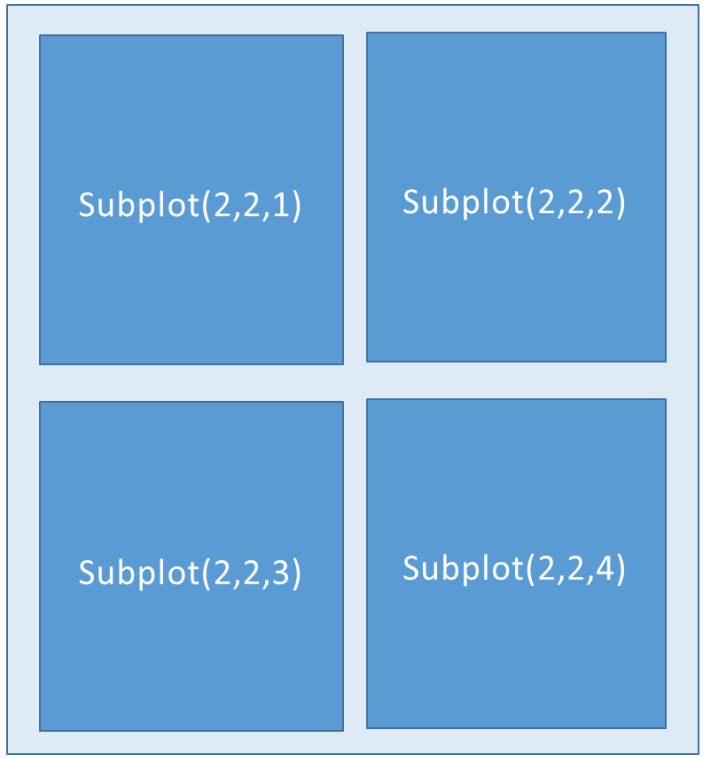
- With subplots, you can arrange plots in a regular grid.

Grid divided into two vertically stacked plots





Grid divided into four plots

Summary

- **Matplotlib** is a widely used Python library for creating static, animated, and interactive visualizations for data analysis.
- It allows users to generate various charts such as **line plots, bar charts, histograms, scatter plots, and heatmaps** for exploring data patterns.
- Matplotlib provides **high customization control over plots**, including axes, labels, colors, legends, and layouts for clear data presentation.

Any Query

Thank You

Happy learning

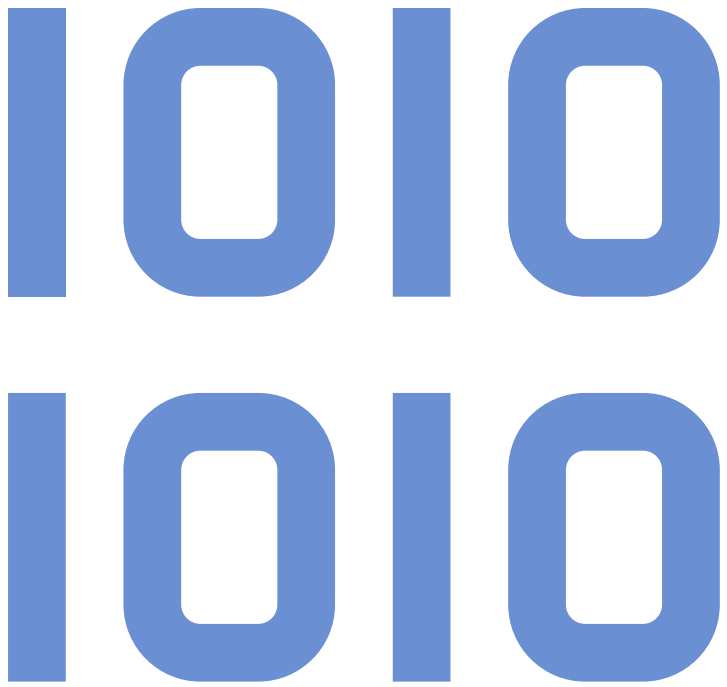
Data Science
Seaborn & Plotly

Presented By : Harsh Dalal

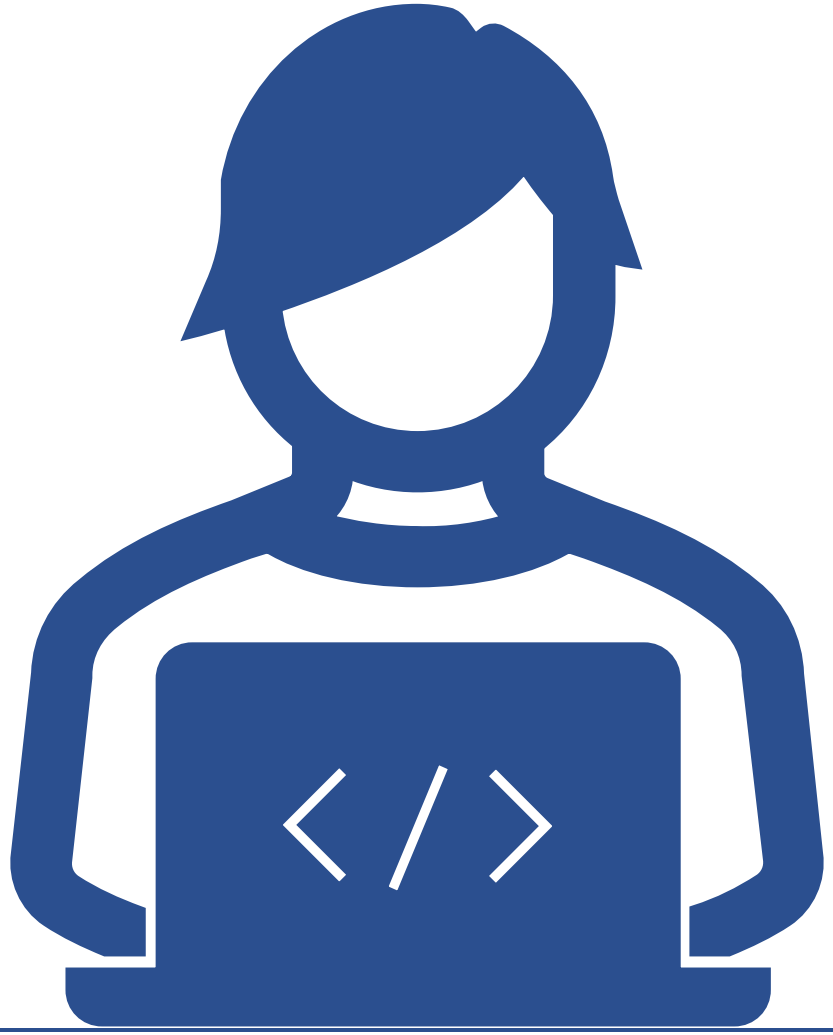
We Make You Trust Data

Agenda

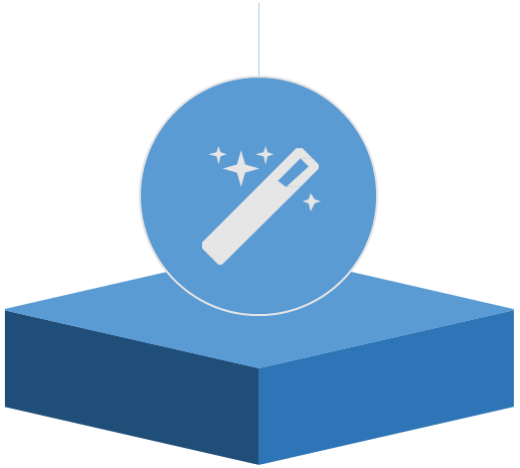
- **Seaborn** is a high-level Python visualization library built on top of **Matplotlib**, mainly used for creating statistical visualizations such as heatmaps, box plots, violin plots, and pair plots.
- It provides attractive default styles and simplifies the visualization of **data distributions, relationships, and correlations** in datasets.
- **Plotly** is an interactive visualization library that enables creation of dynamic charts such as interactive scatter plots, maps, dashboards, and 3D visualizations.
- It is widely used in modern data analytics to build **interactive dashboards and web-based data exploration tools** for better decision-making.



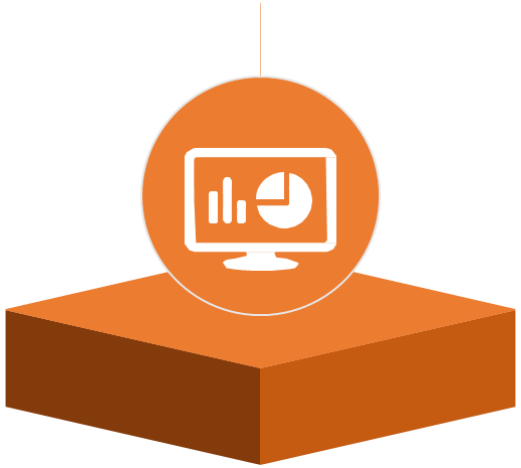
Hands - on



Possesses built-in themes for better visualizations



Has built-in statistical functions which reveal hidden patterns in the dataset

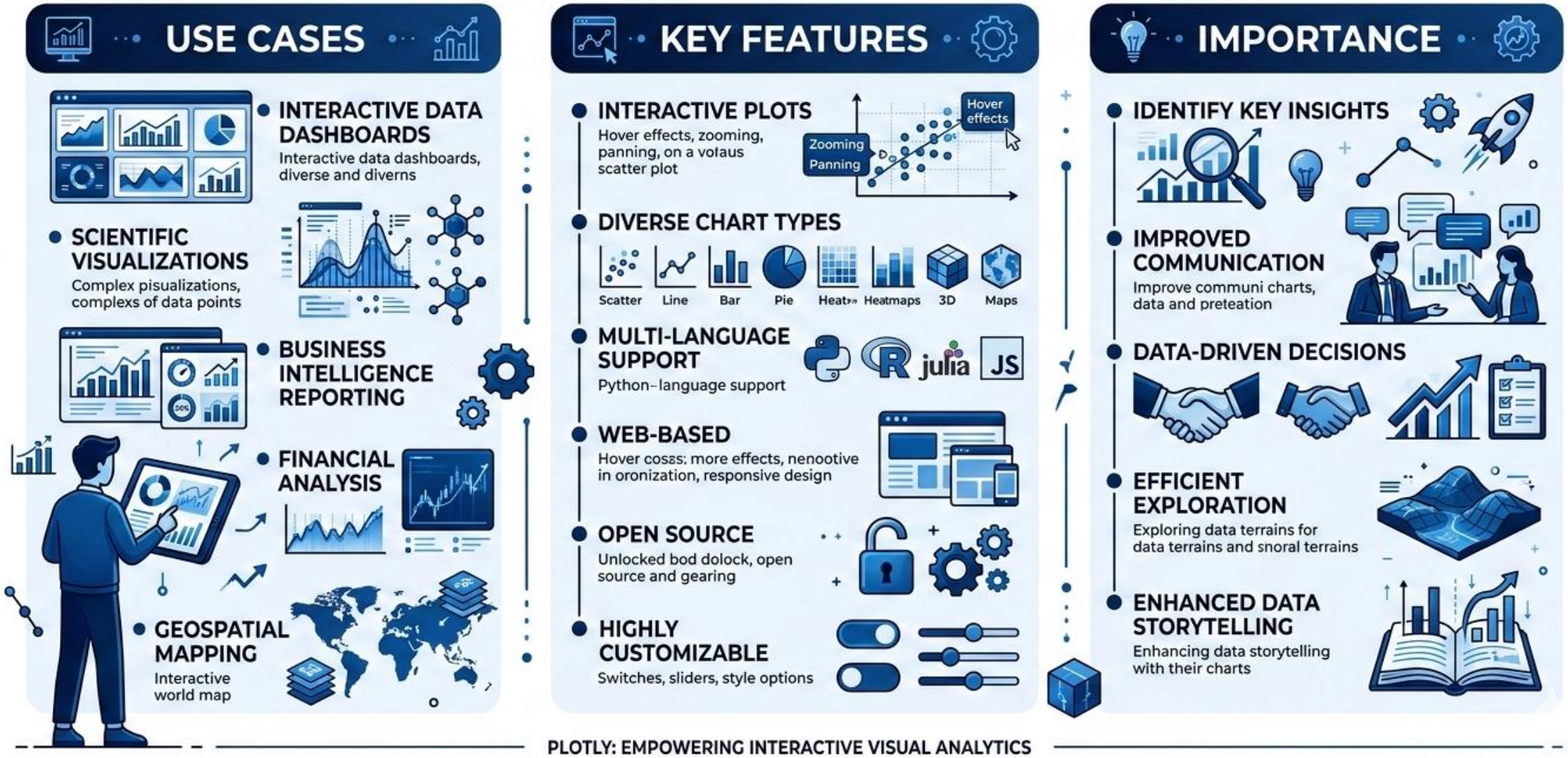


Has functions to visualize matrices of data

Seaborn

- Seaborn is a Python visualization library based on matplotlib. It provides a high-level interface to draw attractive statistical graphics.
- There are several advantages:

Plotly


USE CASES
INTERACTIVE DATA DASHBOARDS
Interactive data dashboards, diverse and diverns
SCIENTIFIC VISUALIZATIONS
Complex pisualizations, complexs of data points
BUSINESS INTELLIGENCE REPORTING
FINANCIAL ANALYSIS
GEOSPATIAL MAPPING
Interactive world map
KEY FEATURES
INTERACTIVE PLOTS
Hover effects, zooming, panning, on a voiaus scatter plot
Zooming
Panning
Hover effects
DIVERSE CHART TYPES
Scatter
Line
Bar
Pie
Heatmaps
3D
Maps
MULTI-LANGUAGE SUPPORT
Python-language support
julia
JS
WEB-BASED
Hover cosas: more effects, nenootive in oronization, responsive design
OPEN SOURCE
Unlocked bod dolock, open source and gearing
HIGHLY CUSTOMIZABLE
Switches, sliders, style options
IMPORTANCE
IDENTIFY KEY INSIGHTS
IMPROVED COMMUNICATION
Improve communi charts, data and preteation
DATA-DRIVEN DECISIONS
EFFICIENT EXPLORATION
Exploring data terrains for data terrains and snoral terrains
ENHANCED DATA STORYTELLING
Enhancing data storytelling with their charts
PLOTLY: EMPOWERING INTERACTIVE VISUAL ANALYTICS

Summary

- **Seaborn** is a Python visualization library built on top of **Matplotlib**, designed for creating clear and attractive statistical graphics.
- It simplifies the visualization of **data distributions, relationships, and correlations** using plots like heatmaps, box plots, violin plots, and pair plots.
- **Plotly** enables creation of **interactive and dynamic visualizations** such as dashboards, maps, and interactive charts for deeper data exploration and presentation.

BANKING LOAN RISK ANALYSIS USING DATA SCIENCE
Predictive Learning Metrics
Probability of Default
Credit Risk Scoring
Portfolio Loss Scoring
Credit Risk Scoring
Portfolio Loss Distribution

Use -cases

Any Query

Thank You

Happy learning